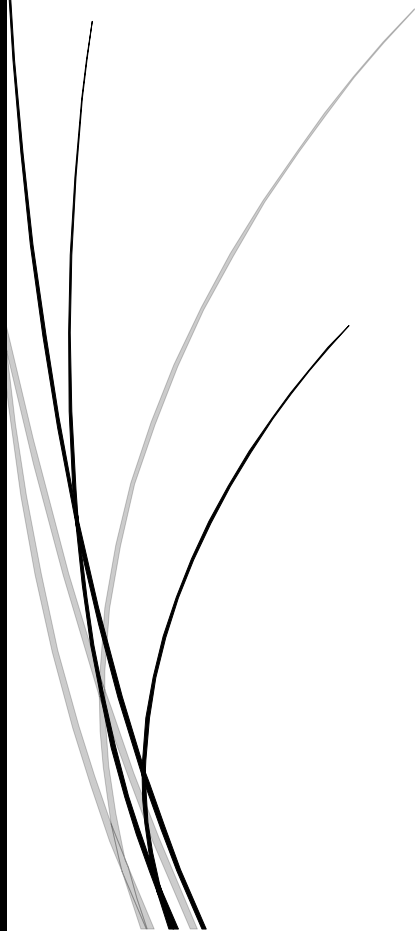
10/30/2026

Assignment-9



Prepare By
~pintu mer

1. Use Wireshark to capture network traffic while you load www.spotify.com in your browser, then identify and note down at least two DNS packets and two HTTP or HTTPS packets in the capture.

Wireshark Traffic Capture – Spotify Website

I used Wireshark to capture network traffic while opening www.spotify.com in my browser.

1. DNS Packets:

- Packet 1: DNS Query – Requested the IP address of www.spotify.com.
- Packet 2: DNS Response – Received the IP address from the DNS server.

2. HTTPS Packets:

- Packet 1: Client Hello – The browser started a secure connection with the server.
- Packet 2: Server Hello – The server responded to establish a secure connection.

Conclusion: Wireshark helps us capture and analyze DNS requests and secure HTTPS communication between a browser and a website.

2. Simulate a DNS lookup for www.flipkart.com using the nslookup command in your terminal and write down the IP address returned along with a brief explanation of what this result means.

   DNS Lookup Using nslookup

   Command: nslookup www.flipkart.com

   IP Address: 10.150.234.131

   Explanation:

   The nslookup command is used to find the IP address of a domain name. DNS converts the domain name www.flipkart.com into an IP address so that the browser can connect to the website.

3. Explain the difference between HTTP and HTTPS by listing 3 key differences, and give a real-world example of an app or website you use that relies on HTTPS for secure communication.

Difference Between HTTP and HTTPS

HTTP	**HTTPS**
1. HTTP stands for Hypertext Transfer Protocol.	HTTPS stands for Hypertext Transfer Protocol Secure.
2. It is less secure.	It is more secure.
3. Data is not encrypted.	Data is encrypted using SSL/TLS.
4. It uses port 80 by default.	It uses port 443 by default.
5. It does not provide secure communication.	It provides secure communication.

Real-world Examples

- HTTP: A basic website that does not use encryption.
- HTTPS: Flipkart (https://www.flipkart.com) uses HTTPS to protect users' login details, personal information and payment data.

4. Open Wireshark and filter for DNS protocol traffic. Try searching for a new website (one you haven't visited recently) in your browser and describe the sequence of DNS queries and responses you observe.Hint:Look for packets with 'Standard query' and 'Standard query response' in the Info column.

DNS Query and Response Using Wireshark

I opened Wireshark and applied the dns filter. Then, I opened a new website (www.wikipedia.org) in my browser.

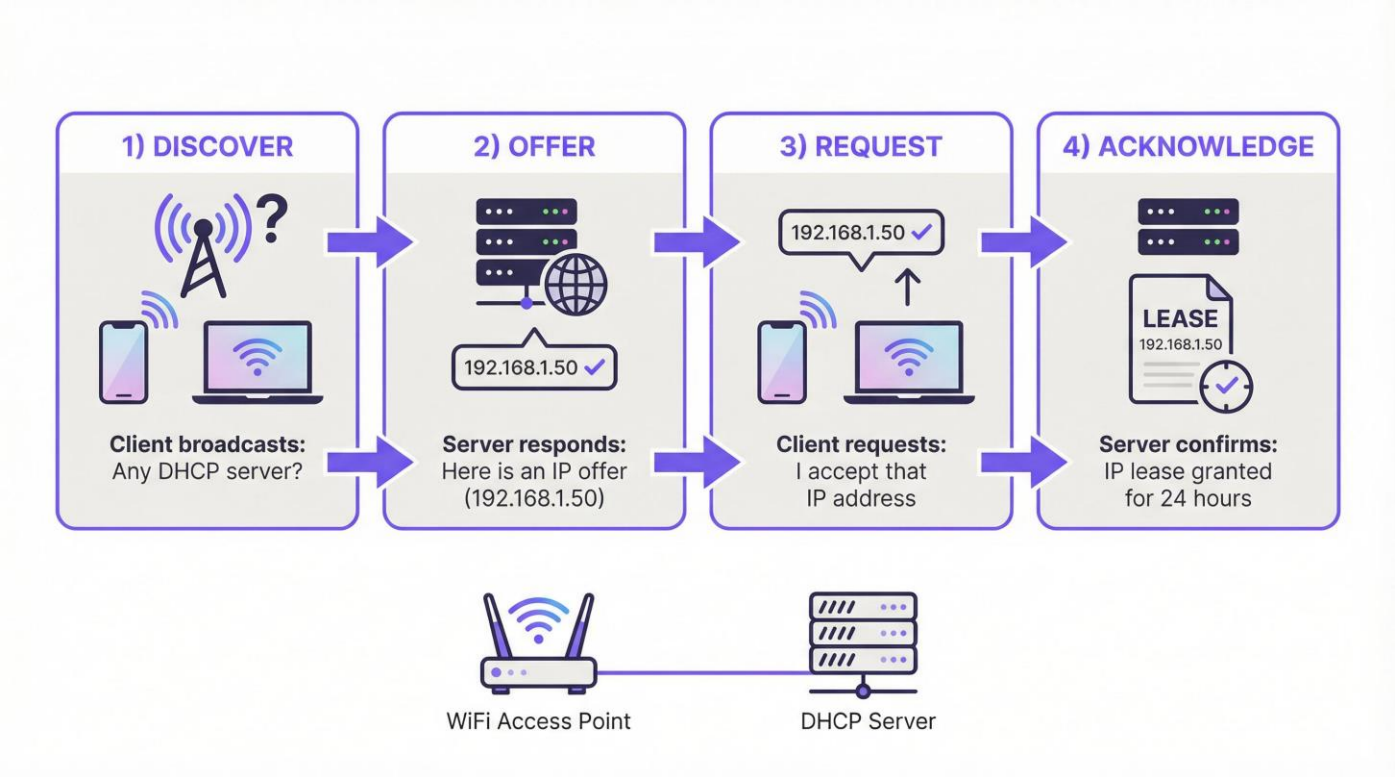
Sequence of DNS Queries and Responses:

1. Standard Query: My computer sent a DNS query to find the IP address of www.wikipedia.org.
2. Standard Query Response: The DNS server replied with the IP address of the website.

3. My computer used the received IP address to connect to the website.

Conclusion: DNS converts domain names into IP addresses and helps computers find websites on the internet.

5. Pick any one protocol from DNS, DHCP, FTP, or SNMP (other than HTTP/HTTPS) and write a short scenario describing how it would be used in a real-world app or service you use (for example, how DHCP helps your phone connect to WiFi when you open Instagram at a cafe).



Protocol: DHCP (Dynamic Host Configuration Protocol)

Real-world Scenario:

When I connect my mobile phone to a cafe's Wi-Fi, the DHCP server automatically assigns an IP address to my phone. It also provides the subnet mask, default gateway, and DNS server. This allows my phone to connect to the internet and use apps like Instagram.

Conclusion: DHCP automatically provides network settings to devices, making network connectivity easier.